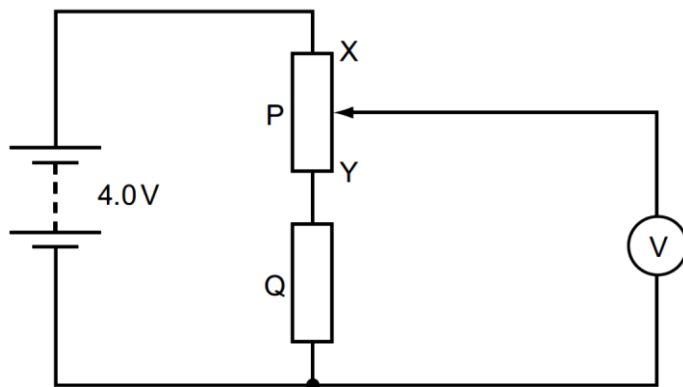
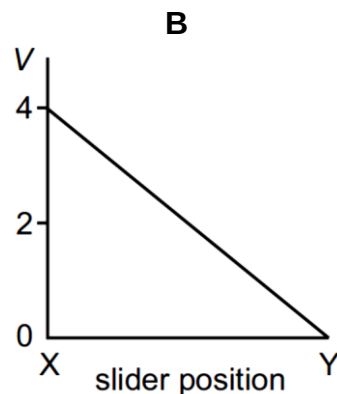
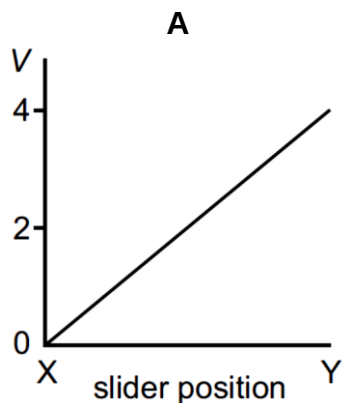


- 21** In the circuit below, P is a potentiometer of total resistance $10\ \Omega$ and Q is a fixed resistor of resistance $10\ \Omega$. The battery has an e.m.f. of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance. The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

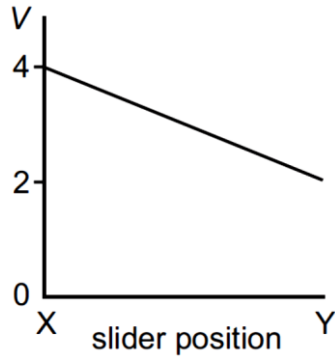
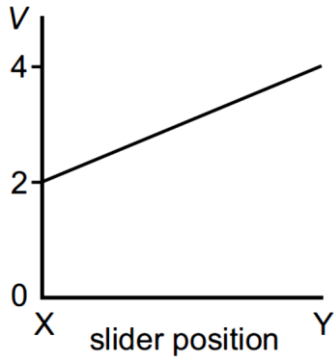


Which of the following shows the correct graph obtained?



C

D



At X, voltmeter reads emf of supply = 4 V

D At Y, voltmeter reads pd across resistor Q = 2 V